Blocking and Schur bounds: the surface-to-gap obstruction

A block criterion based on the norm of straddling plaquettes divided by an intra-block gap has a surface cost. For M^3-site blocks of a three-dimensional spatial lattice the counted sufficient bound is

$$\beta_{\mathrm{block}} = \frac{24M^2N}{g^2\delta(g;M)}, \qquad \Delta_M = \frac{\hbar c}{a}\delta(g;M).$$

Its strong-coupling asymptotic exceeds the direct criterion by $3M^2/8$, and a small-volume zero-mode model gives M^3 scaling. These are regime-dependent comparisons, not a proof of monotonicity at every coupling. The inequality $\beta_{\mathrm{block}}(M) \geq \beta_{\mathrm{block}}(1)$ follows if $\delta(g;M) \leq M^2\delta(g;1)$; a separate upper bound on $\delta(g;M)$ does not establish this comparison. The original all-coupling claim is withdrawn. No uniform physical mass gap or continuum construction follows.

1. Blocking, norms and the Schur issue

Partition a periodic lattice of $N_s = MK$ sites per side into K^3 blocks, assign links by midpoint and write

$$H = \sum_\alpha H_\alpha + \sum_{p \text{ straddling}} w_p, \qquad H_\alpha = \frac{\hbar c g^2}{2a}\sum_{\ell\in\alpha}(-\Delta_\ell) + \sum_{p\subset\alpha} w_p.$$

The earlier calculation used $6M^2$ straddling plaquettes per block; face-crossing counting may require a factor of two unless terms are half-attributed. The displayed numerical criterion is conditional on that assignment. For each plaquette, $\|w_p\| \leq 4N\hbar c/(ag^2)$. Edge counting and interaction assignment must be fixed before applying a particular polymer theorem. The block Hamiltonians have finite-volume ground states and gap $\Delta_M > 0$. A product projection P onto block low-energy subspaces gives a compression $A = PHP$; a Schur reduction additionally requires a lower bound on the complement $D = \bar{P}H\bar{P}$, including the straddling interactions. Such a bound does not follow merely from the uncoupled block gap.

If $B = PH\bar{P}$ and a complementary resolvent bound is supplied, a naive per-block estimate is

$$\frac{\|B\|^2}{\Delta_M} \lesssim 36M^4\frac{\hbar c}{a}\frac{16N^2}{g^4\delta(g;M)}.$$

This measures a large boundary contribution as well as level-dependent variation. The relative Schur lemmas in the Feshbach note show why a common energy shift can cancel from a gap estimate, while its energy dependence and residual variation still need control. A proposed bound

$$\|R(E)\| \leq C\sum_{p \text{ straddling}}\frac{\|w_p\|^2}{\Delta_M}$$

would replace the naive M^4 counting by M^2, but is a hypothesis: polymer connectedness by itself does not prove this particular Schur operator inequality or its volume-uniform constants. The earlier one-step threshold and induction based on that unproved inference are withdrawn.

2. Direct block criterion and its limitations

Normalize a block Hamiltonian by its gap, $h_\alpha = (H_\alpha - E_0^\alpha)/\Delta_M$, and bound the normalized inter-block perturbation by β_{block} above if that count and assignment hold. Yarotsky's bounded-perturbation theorem gives a sufficient criterion of the form $\beta_{\mathrm{block}} \leq \beta_*$, subject to its product-space, unique block-ground-state, finite-range and interaction-assignment hypotheses. Its polymer proof (Commun. Math. Phys. 261 (2006), §2, passage-level reading) has connected excited-region weights; it can be applied only after those hypotheses are checked, and it is not the asserted Schur variation estimate. See the group-specific application and its scope in strong coupling.

At strong coupling, in a block containing an elementary loop and in the gauge-invariant sector, the electric excitation is a fundamental plaquette loop with energy $2C_2(R_{\min})g^2\hbar c/a$. For fixed block geometry as $g \to \infty$ this gives

$$\beta_{\text{block}} \sim \frac{12M^2N}{C_2 g^4}, \qquad \frac{\beta_{\text{block}}}{\beta_{\text{direct}}} \sim \frac{3M^2}{8}, \qquad \beta_{\text{direct}} = \frac{32N}{C_2 g^4}.$$

Thus the displayed sufficient estimates compare unfavourably at $M = 2$ in that asymptotic regime; they are bounds, not exact optimal thresholds. Boundary conditions and the sector of the block operator matter.

For a periodic small-volume zero-mode model, the conditional asymptotic $\Delta_M \sim \delta_1 g^{2/3}\hbar c/(Ma)$ would give

$$\beta_{\text{block}} \sim \frac{24M^3N}{\delta_1 g^{8/3}}.$$

It must not be imported without proof into the free-boundary blocks above or treated as a uniform fixed-cutoff theorem; see the Feshbach reduction. Algebraically the criterion is no better than its $M = 1$ analogue if $\delta(g; M) \leq M^2\delta(g; 1)$, assuming both are defined with compatible boundary conventions. This condition has not been established for all couplings. A one-sided upper bound such as the Polyakov variational bound cannot supply the missing lower comparison with $\delta(g; 1)$.

3. Running coupling and generated interactions

The one-loop count for successive doublings, conditional on the running law $g_{n+1}^{-2} = g_n^{-2} - 2b_0 \log 2$ with $b_0 = 11N/(48\pi^2)$, is

$$n \simeq \frac{g_{\text{UV}}^{-2} - g_{\text{thr}}^{-2}}{2b_0 \log 2}.$$

For $SU(2)$, $2b_0 \log 2 = 11 \log 2/(12\pi^2) \approx 0.064$; the illustrative inputs $g_{\text{UV}}^2 = 1/2$ and $g_{\text{thr}}^2 \approx 20$ yield about thirty doublings. This perturbative extrapolation is not a certified arrival at a strong-coupling domain. The number increases with the ultraviolet cutoff along an asymptotically free trajectory. A projection generates longer-range and other interactions, and a coupling label alone does not prove an iterated effective-action or gap estimate. An RG construction must control those generated terms and observable matching at each scale.

4. Consequence for STATE

Keep the boundary count, conditional comparison, asymptotic strong-coupling calibration and Schur distinction. Neither a general no-improvement theorem nor a connected-Schur estimate is established. Control of the blocked action and gap through the weak-to-intermediate regime remains open; the physical energy unit is $\hbar c/a$ and the volume and continuum limits are separate obligations.